Canada EV Charging Infrastructure Analysis

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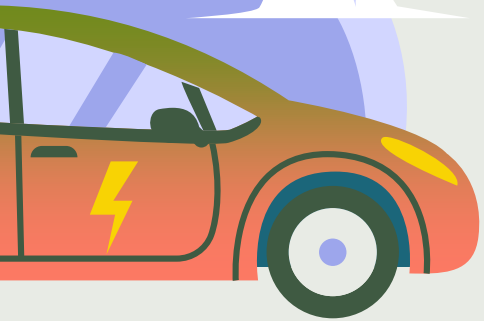
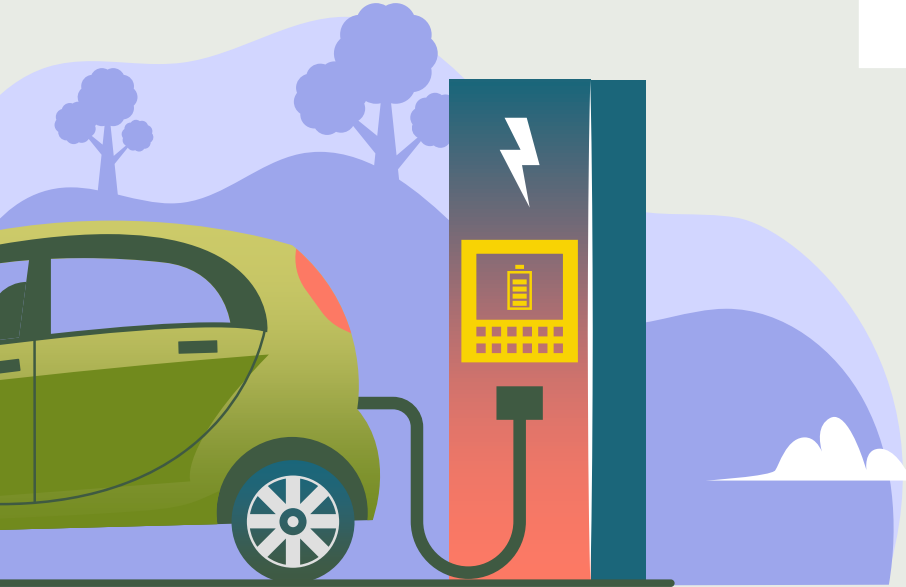


01

Business Understanding



Client Background & Problem

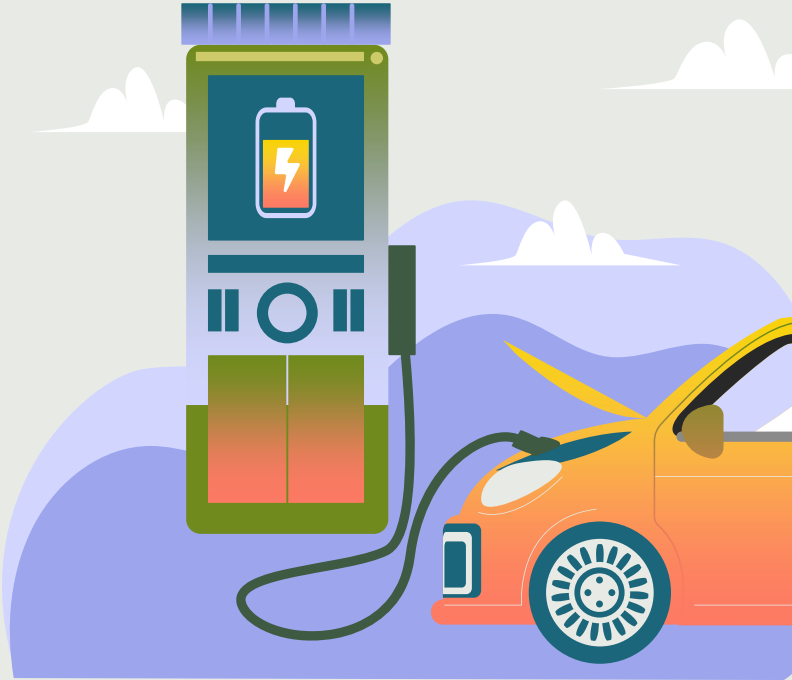
Transportation Canada Targets

- 20% by 2026
- 60% by 2030
- 100% by 2035

EV charging accessibility

Charging infrastructure gaps

EV adoption declining



Client & Stakeholders

Primary Client

Natural Resources Canada (NRCan)

- Meeting EV adoption goals
- Charging infrastructure allocation

Key Stakeholders

City Planners

- Planning future areas

EV Manufacturers

- Aligns with EV usage changes

Energy Providers

- Maintain grid stability

Citizens

- Charging accessibility
- Purchasing incentives

Project Objective

Objective

Provide recommendations to identify where EV charging infrastructure needs improvements

Limitations

Plan involves zero-emission vehicles (ZEV)

- Analysis focused on electric vehicles (EV)

Business Questions

01

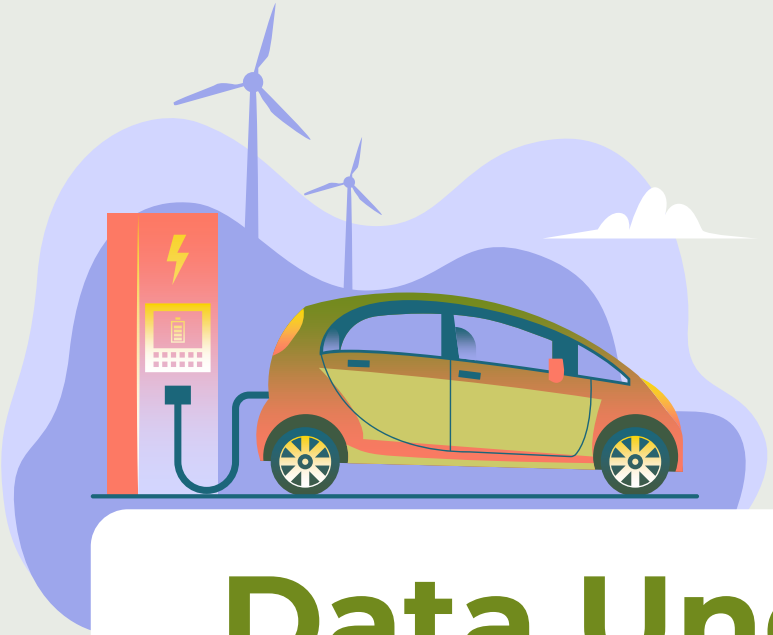
Where should EV charging investments be prioritized?

02

How many new chargers are required under different adoption scenarios

03

Which charger type delivers greater value per installation



02

Data Understanding

Census 2024

Population Density

Alt Fuel Station

Alternative Fueling Station Info

Highway System

Geographical highway information

EV data explorer

EV Sales

Charging Stations

Charging Station Info

Census 2024

Population Density

Information

Size:

- 8 KB
- 3 Columns
- 294 Rows

Population information for different regions across Canada

	A	B	C	D	E	F	G
1	Code	Area Name	Population				
2	1001	Division No. 1	300445				
3	1002	Division No. 2	19114				
4	1003	Division No. 3	13609				
5	1004	Division No. 4	19917				
6	1005	Division No. 5	42228				
7	1006	Division No. 6	38947				
8	1007	Division No. 7	33951				
9	1008	Division No. 8	34333				
10	1009	Division No. 9	14551				
11	1010	Division No. 10	25460				
12	1011	Division No. 11	2692				
13	1101	Kings	20633				
14	1102	Queens	106078				
15	1103	Prince	51839				
16	1201	Shelburne	14394				
17	1202	Yarmouth	26264				

Census 2024

Population Density

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Alt Fuel Station

Alternative Fueling Station Info

Information

Size:

- 98,255 KB
- 85 Columns
- 265,718 Rows

Information on alternative fueling stations such as location, connector type, and facility type

	A	B	C	D	E	F	G	H	I
1	Fuel Type Code	City	Latitude	Longitude	EV Connector Types	Country	Facility Type	EV J1772 Connector Count	EV J1772 Power Output (kW)
2	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
3	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
4	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
5	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
6	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
7	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
8	ELEC	Los Angeles	34.040539	-118.271387	J1772	US	PARKING_GARAGE	1	
9	ELEC	La Jolla	32.89947	-117.243	J1772	US	HOSPITAL	1	
10	ELEC	Sepulveda	34.221665	-118.468371	J1772	US	CAR_DEALER	1	
11	ELEC	Sepulveda	34.221665	-118.468371	J1772	US	CAR_DEALER	1	
12	ELEC	Riverside	33.909914	-117.459053	J1772	US	SHOPPING_MALL	1	
13	ELEC	Riverside	33.909914	-117.459053	J1772	US	SHOPPING_MALL	1	
14	ELEC	Riverside	33.909914	-117.459053	J1772	US	SHOPPING_MALL	1	
15	ELEC	Riverside	33.909914	-117.459053	J1772	US	SHOPPING_MALL	1	
16	ELEC	Pasadena	34.14762	-118.147111	J1772	US	PARKING_GARAGE	26	
17	ELEC	Pasadena	34.145138	-118.152655	J1772	US	PARKING_GARAGE	1	
18	ELEC	Pasadena	34.145138	-118.152655	J1772	US	PARKING_GARAGE	1	
19	ELEC	Pasadena	34.145138	-118.152655	J1772	US	PARKING_GARAGE	1	
20	ELEC	Pasadena	34.145119	-118.150133	J1772	US	PARKING_GARAGE	1	

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Highway System

Geographical highway information

Information

Size:
- 36,400 KB

GDB file type
- Contains geographical points and line information

Gives geographical information of different highway systems

A collection of points and lines to provide visual information on highway location and outline

Census 2024

Population Density

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EV sales

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Charging Station Info

EV data explorer

EV sales

Information

Size:
- 915 KB

An excel worksheet showing global EV sales from 2010 to 2024

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
7	Sum of value	year														
8	region_country	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
9	Australia		49	170	190	370	760	670	1,200	1,800	6,300	5,200	17,000	33,000	87,000	91,000
10	Austria				650	1,700	1,700	3,800	5,400	6,800	9,300	16,000	33,000	34,000	48,000	45,000
11	Belgium	48	280	570	500	1,300	1,400	2,100	2,700	3,600	8,800	15,000	23,000	38,000	93,000	130,000
12	Brazil	10	8	22	39	61	61	130	140	180	540	800	2,900	8,500	19,000	62,000
13	Bulgaria						21	5	68	190	180	140	430	1,000	1,800	1,700
14	Canada		220	620	1,600	2,800	4,400	5,200	8,700	23,000	32,000	37,000	59,000	91,000	130,000	190,000
15	Central and South America	10	15	53	66	93	110	210	390	810	2,000	2,400	7,200	18,000	36,000	98,000
16	Chile		6	5	5	5	13	22	120	130	220	160	560	1,300	1,600	4,500
17	China	1,100	4,800	9,600	15,000	49,000	150,000	260,000	470,000	820,000	830,000	920,000	2,700,000	4,400,000	5,400,000	6,400,000
18	Colombia		1	26	22	27	39	28	92	250	760	670	840	1,900	3,600	11,000
19	Costa Rica							22	14	230	350	590	1,000	2,600	5,800	11,000
20	Croatia										190	530	1,400	1,400	1,600	1,800
21	Cyprus										23	42	84	400	790	1,200
22	Czech Republic						300	200	390	700	760	3,300	2,700	3,900	6,700	11,000
23	Denmark		420	490	500	1,600	4,300	1,300	710	1,500	5,500	14,000	25,000	31,000	63,000	89,000
24	Estonia						34	35	26	85	80	340	480	730	1,400	1,300
25	Finland		30	51	50	180	240	220	500	780	1,900	4,200	10,000	15,000	30,000	22,000
26	France	190	2,600	5,700	8,900	11,000	18,000	23,000	26,000	33,000	45,000	110,000	170,000	210,000	310,000	300,000

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Charging Station Info

Charging Stations

Charging Station Info

Information

Size:

- 22,965 KB
- 11 Columns
- 242,418 Rows

Global information on charging stations including location, type, and output

	A	B	C	D	E	F	G	H	I	J	K
1	id	name	city	country_code	state_province	latitude	longitude	ports	power_kw	power_class	is_fast_dc
2	307660	Av. de Tarragona	Andorra	AD		42.505254	1.528861	10	300	DC_ULTRA_(>=150kW)	TRUE
3	301207	Parquing Costa Rodona	Encamp	AD		42.537213	1.727014	10	22	AC_HIGH_(22-49kW)	FALSE
4	301206	Hotel Naudi		AD		42.576811	1.666061	1	11	AC_L2_(7.5-21kW)	FALSE
5	301205	Hotel Piolets Soldeu Centre		AD		42.576466	1.667317	1	22	AC_HIGH_(22-49kW)	FALSE
6	301204	Hotel Serras		AD		42.579458	1.659215	3	11	AC_L2_(7.5-21kW)	FALSE
7	301203	HOTEL LLOP GRIS		AD		42.57807	1.64882	2	22	AC_HIGH_(22-49kW)	FALSE
8	301202	Aparcament font del ferro		AD		42.58257	1.636854	3	11	AC_L2_(7.5-21kW)	FALSE
9	301201	Hotel l'Ermita		AD		42.554031	1.589863	1	11	AC_L2_(7.5-21kW)	FALSE
10	301200	Restaurante Hotel Les Pardines	Encamp	AD		42.530853	1.600491	1	11	AC_L2_(7.5-21kW)	FALSE
11	301199	La Solana Apartaments & Spa	Encamp	AD		42.534027	1.588139	1	7.4	AC_L1_(<7.5kW)	FALSE
12	301198	Estacio Ordino Arcalis Planells		AD		42.629215	1.498094	6	7.4	AC_L1_(<7.5kW)	FALSE
13	301197	Hotel Ordino	Ordino	AD		42.5557	1.533428	1	3.3	AC_L1_(<7.5kW)	FALSE
14	301196	Hotel Princesa Parc	La Massana	AD		42.573917	1.482224	1	11	AC_L2_(7.5-21kW)	FALSE
15	301195	Parking la Borda del avi	La Massana	AD		42.551185	1.510482	4	11	AC_L2_(7.5-21kW)	FALSE
16	301194	Apartaments Giberga	La Massana	AD		42.54662	1.524825	1	7.4	AC_L1_(<7.5kW)	FALSE
17	301193	AnyÃ³sPark The Mountain & Well	La Massana	AD		42.531792	1.524056	4	7.4	AC_L1_(<7.5kW)	FALSE
18	301192	Hotel Silken Insitu Eurotel	les Escaldes	AD		42.513697	1.533125	1	11	AC_L2_(7.5-21kW)	FALSE
19	301191	Hotel Panorama	les Escaldes	AD		42.507949	1.540879	2	11	AC_L2_(7.5-21kW)	FALSE
20	301190	Hotel Metropolis	les Escaldes	AD		42.510163	1.540748	2	11	AC_L2_(7.5-21kW)	FALSE

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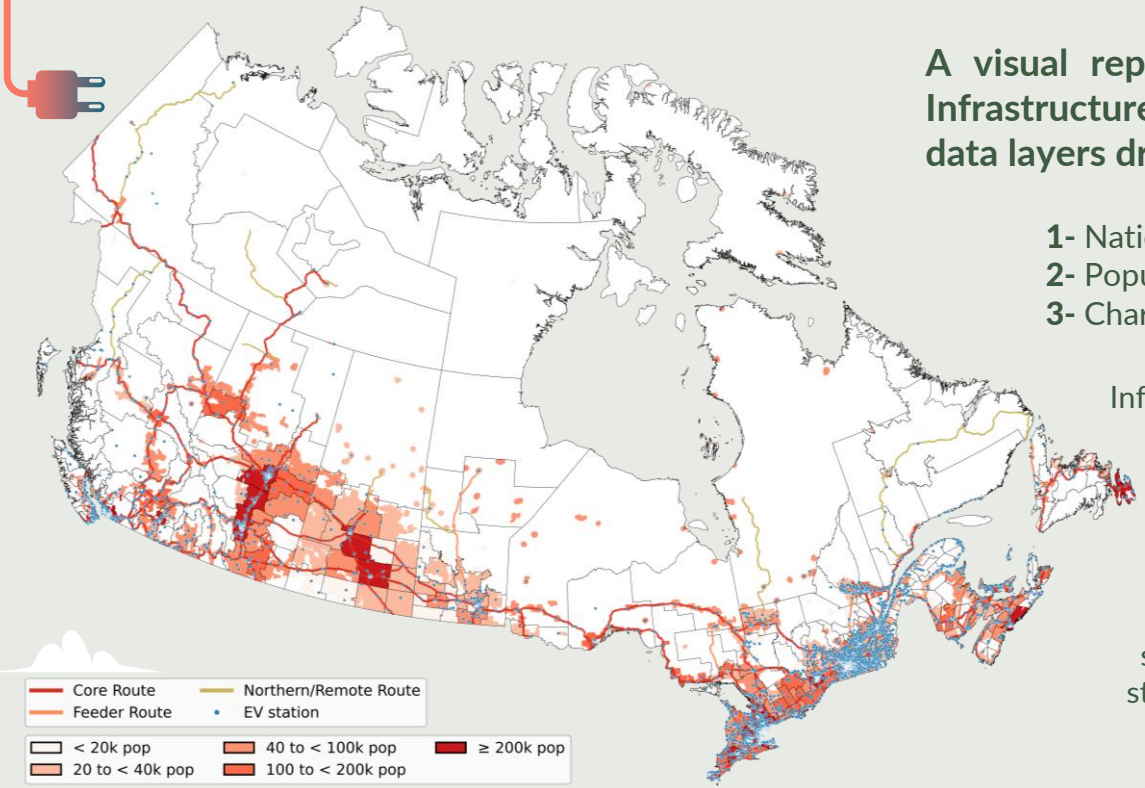
Data Dictionary

Field	Data Type	Description
City	String	Provides geographical information for location
EV Network	String	Shows if networked through Chargepoint or Tesla etc.
Longitude/latitude	Float	Coordinates to support spatial visualizations
Open Date	Date	When the station opened to show growth
EV Connector Types	String	Provides information on connector type to highlight standardization

03

Modeling

Federal Infrastructure Gaps

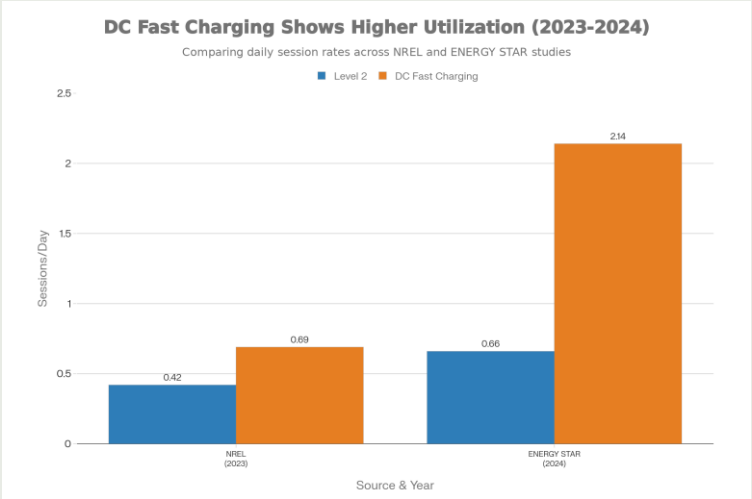


A visual representation of where the Charging Infrastructure is lacking, built with 3 data different data layers drawn on top of the Canadian map:

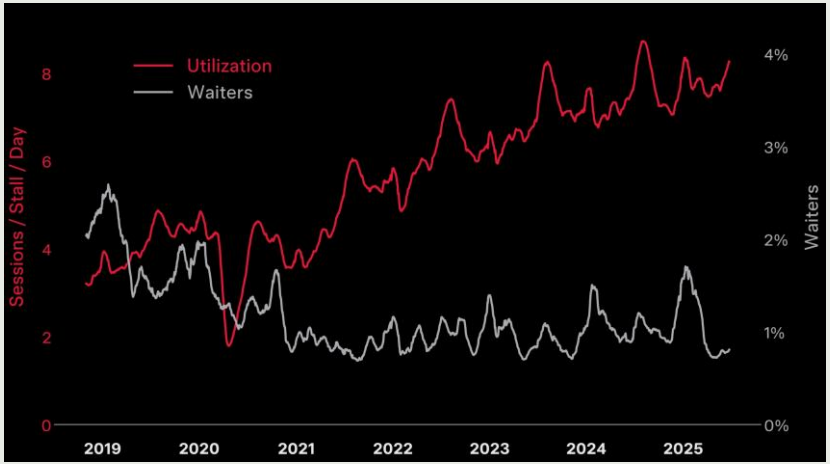
1- National Highway System hierarchically colored
2- Population density with ecumene filter
3- Charging Station locations

Infrastructure is heavily concentrated in Québec, Ontario, specially along the Québec–Windsor corridor, and BC. All other regions show significant population centers with minimal charger coverage, creating gaps that threaten long-distance EV travel and regional adoption. Québec's 35 EV-to-charger ratio contrasts sharply with under-served provinces, requiring strategic federal investment to balance national coverage.

Comparison Score Level 2 vs DCFC



Sessions per day for Level 2 (slow) chargers. (Sources: in chart)

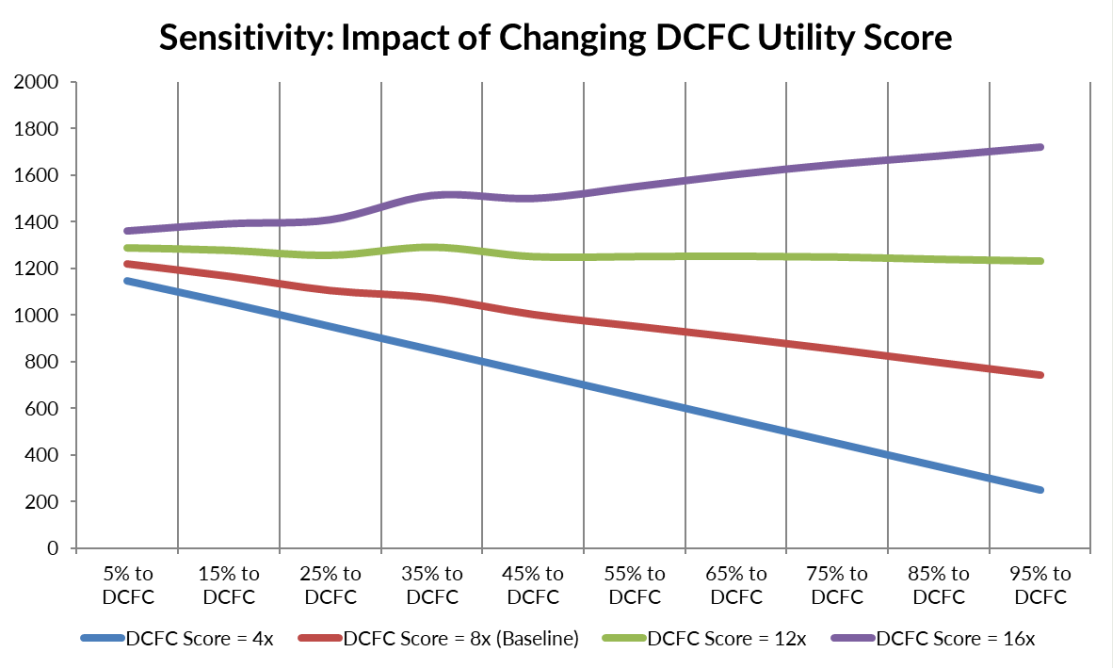


Tesla Supercharging network: utilization and waiters. (Source: Tesla)

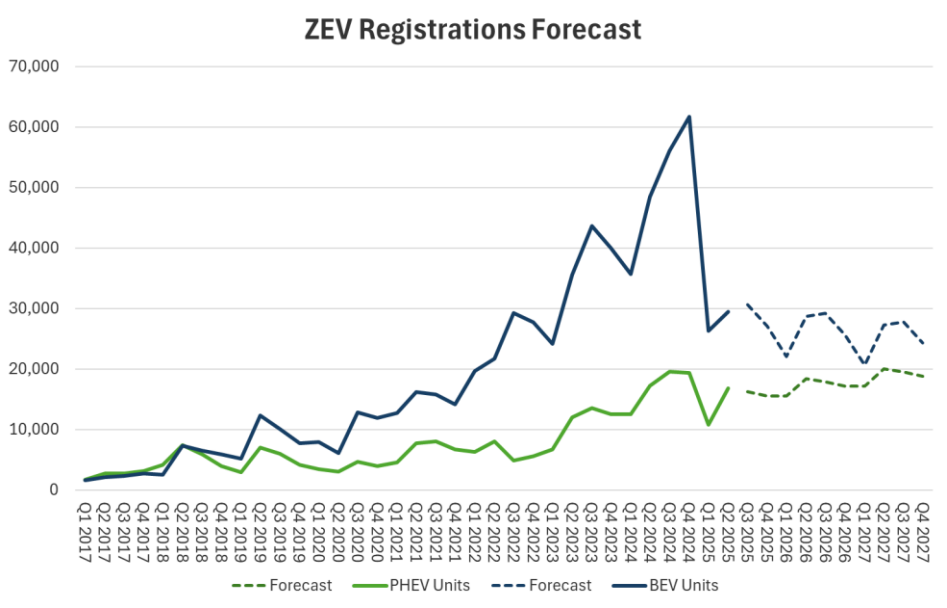
Monte Carlo Simulation

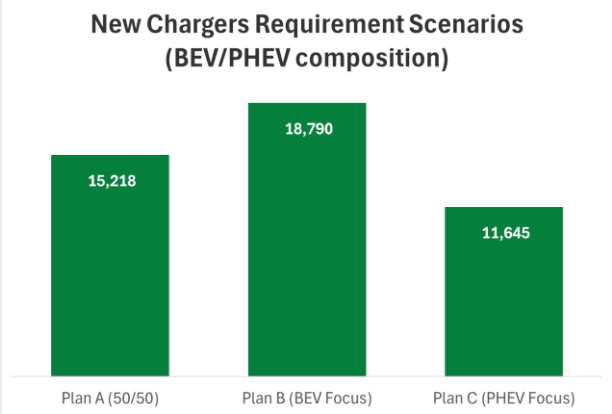
Monte Carlo Simulation Parameters:

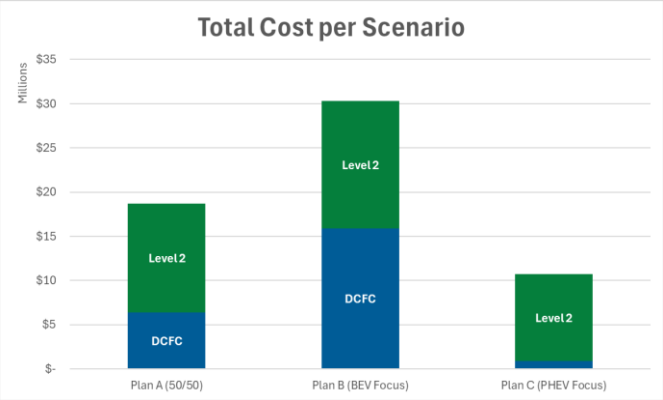
1,000 runs, $100M budget allocated between Level 2 ($5k–$12k) and DCFC ($80k–$150k) chargers. Utility score modeled as DCFC = 4×, 8× (baseline), 12×, or 16× Level 2 value. Each point represents average utility across simulations for a given allocation mix.



How Many Chargers We Need







Interpretation of Results

Canadian EV owners "overwhelmingly satisfied" with EVs, access to charging a key purchase factor

EV Charging Infrastructure Importance

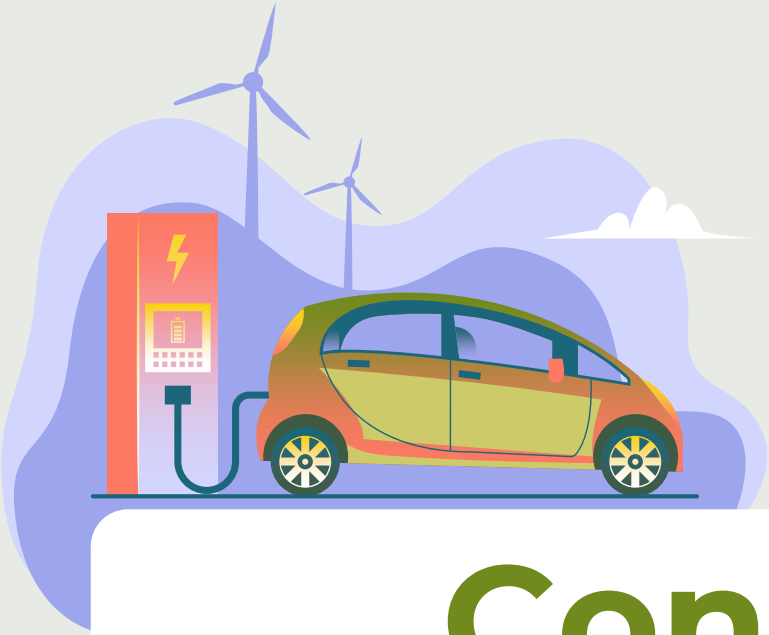
It is essential for the long-term sustainability of the project that proper charging infrastructure is built to sustain the growing market for electric vehicles and strengthen consumer trust on the new technology, then by consequence increasing its sales.

Short Term or Long Term

Choosing to build slow chargers would result in better short-term impact and benefits, but would also incentivize the market for PHEVs, which does not fully comply with the plans end goal of 100% zero emissions new vehicles sales.

"46% of EV owners in Québec find station availability adequate, compared to 28% in Ontario and 21% in B.C."

Source: Electric Autonomy Canada



04

Conclusion

Conclusion

- **Infrastructure Gaps:** Canada has 35,000 chargers but needs **100,520** to meet policy goals; urban, highway, and provincial disparities exist.

- **Demand Forecasting:** Depending on fleet composition, Canada will require **679,000** public charging ports by 2040.

  -> High-BEV scenario: more fast chargers needed for long-range trips.
  -> High-PHEV scenario: lower public charging demand.

- **Cost-Effectiveness:** Multifamily retrofits serve more vehicles per dollar than highway DCFCs, but DCFCs are essential for long trips.

- **Strategic Impact:** Evidence-based allocation ensures funds maximize value, target high-priority regions, and support achieving **20% ZEV sales by 2026, 60% by 2030, 100% by 2035**.

Recommendations

- **Prioritize Underserved Areas:** Target regions with the largest gaps—urban neighborhoods, rural provinces, and incomplete highway corridors to address the current shortfall of **35,000 vs 100,520 needed chargers**.

- **Plan by Fleet Composition:** Allocate chargers based on projected scenarios high-BEV vs high-PHEV to ensure infrastructure meets **679,000 public ports needed by 2040**.

- **Maximize Cost-Effectiveness:** Focus on multifamily building retrofits for the highest vehicles-served per dollar; deploy DCFCs strategically on highways for long-range trips.

- **Evidence-Based Investment:** Use geospatial, demand, and cost analyses to guide funding allocation, ensuring progress toward **20% ZEV by 2026, 60% by 2030, 100% by 2035**.